Correlation and Rule Evaluation for Phishing Discovery: A Comparative Study of Interestingness Measures

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Aylin Şahin
Computer Engineering Department
TOBB University of Economics and Technology
Ankara, Turkey
Student ID: 221104088 aylinsahin@etu.edu.tr

***Abstract*—Phishing remains a critical cybersecurity challenge, necessitating the identification of recurring patterns within deceptive websites to enhance detection systems. This study investigates the application of Association Rule Mining (ARM) for phishing discovery, focusing on a comparative evaluation of eight different interestingness measures. Utilizing the UCI Phishing Websites Dataset, the performance of Apriori and FP-Growth algorithms was compared. Despite theoretical expectations, the Apriori algorithm achieved significantly faster execution times (0.65 seconds) compared to FP-Growth (91.22 seconds) for the specified support threshold, illustrating the computational overhead of tree construction in high-dimensional data. To identify the most effective criteria for phishing detection, metrics including Support, Confidence, Lift, Chi-Square, Cosine, All-Confidence, Max-Confidence, and Kulczynski were analyzed. A controlled experiment was conducted by adding 10,000 synthetic null transactions to test the null-invariance property of these metrics. The results reveal that traditional measures like Lift and Chi-Square are highly sensitive to dataset size, with Lift values inflating by over 90%, potentially leading to misleading interpretations in imbalanced datasets. In contrast, null-invariant measures such as the Kulczynski measure demonstrated 100% stability, consistently identifying "SSLfinal_State_No_HTTPS" and high "URL_of_Anchor" density as the most robust predictors of phishing activity. This research concludes that null-invariant interestingness measures are essential for extracting trustworthy and scalable security patterns, providing a more reliable foundation for building automated phishing detection systems.**

***Index Terms*—phishing detection, association rule mining, interestingness measures, null-invariance, data mining algorithms**

I. Introduction

A. Problem Definition and Motivation

With the rapid advancement of technology, cyber security and cyber attacks have become a major issue in today's world. Among cyber attacks, phishing is one of the most common. Phishing involves the use of deceptive websites that impersonate legitimate organizations in order to steal sensitive user information, such as login credentials and financial data. Despite advanced classification models, there is a growing need for security models that not only detect threats but also identify the key patterns and characteristics that define a phishing website. Association Rule Mining provides a powerful framework for discovering hidden relationships between website features and their legitimacy. However, the effectiveness of mining is highly dependent on the choice of interestingness measures.

Standard measures like Support and Confidence often fail to reflect the true strength of a correlation, especially in unbalanced cyber security datasets. Furthermore, many traditional correlation measures are sensitive to null transactions. Therefore, evaluating rules using null-invariant measures is crucial for building trustworthy phishing detection systems.

B. Research Question

This study aims to address the following research questions:

Q1: What are the most effective criteria for identifying phishing sites?

Q2: Do the Support and Confidence metrics mislead us when dealing with imbalanced data, such as in phishing detection?

Q3: Which correlation measure provides the most stable result without being affected by an increase in null transactions?

C. Overview of Approach

Data mining was used to investigate these questions. Data pre-processing was carried out on a selected dataset. Subsequently, the Apriori and FP-Growth algorithms were applied to compare their execution times and scalability. The association rules were examined using 8 different metrics. These are Support, Confidence, Lift, Chi-Square, Cosine, All-Confidence, Max-Confidence and Kulczynski. Finally, null transactions were added to the dataset to observe the null-invariance property. Following this addition, changes in all metrics were examined.

II. RELATED WORK

The detection of phishing has evolved from simple blacklisting to sophisticated automated systems utilizing machine

learning and data mining techniques. Since phishing attacks rely on both social engineering and technical deception, researchers have focused on identifying recurring patterns in the structure of malicious URLs and the behavioral characteristics of web pages.

A significant study in this field was conducted by Jeeva, S Carolin and Rajsingh, Elijah Blessing [1]. Jeeva, S Carolin and Rajsingh, Elijah Blessing investigate the use of Apriori and Predictive Apriori algorithms to identify phishing URLs. The authors extracted 14 heuristic features from the PhishTank dataset, such as URL length, the number of slashes, and the presence of Transport Layer Security (TLS). Their findings emphasize that the absence of TLS, the unavailability of Top-Level Domains (TLD), and specific keywords in the URL path are the most reliable indicators of phishing activity. Using the Apriori algorithm, they achieved a 93% detection rate by focusing on rules with high confidence.

While Jeeva, S Carolin and Rajsingh, Elijah Blessing's research [1] and this study both leverage the Apriori algorithm, they diverge significantly in their fundamental objectives and evaluation methodologies. The work in [1] is primarily centered on feature engineering and the discovery of specific URL-based heuristics using traditional support and confidence frameworks. In contrast, this study shifts the focus from feature discovery to the rigorous evaluation of interestingness measures themselves. While Jeeva, S Carolin and Rajsingh, Elijah Blessing's research [1] relies on the Predictive Apriori algorithm to rank rules by accuracy, this research investigates the performance trade-offs between Apriori and FP-Growth, specifically addressing the computational overhead and the "curse of dimensionality" inherent in high-dimensional cybersecurity datasets.

Beyond feature discovery, recent research has focused on improving the speed and efficiency of detection algorithms. Hammed, Mudasiru and Soyemi, Jumoke [2] proposed a hybrid model that combines Association Rule Mining with the Firefly Algorithm to detect phishing attacks on social media platforms. The primary goal of their study was to solve the slow processing time of the standard Apriori algorithm, which often struggles with large datasets. By analyzing 21 URL-based features and using the Firefly Algorithm to optimize the support calculation process, their system achieved a high accuracy of 99.3% while significantly reducing the time needed to classify messages.

The research by Hammed, Mudasiru and Soyemi, Jumoke [2] is closely related to the algorithmic performance analysis conducted in this study. Both research efforts recognize that high-dimensional datasets can lead to memory bottlenecks and slow execution times. However, the approaches to solving this problem differ: while Hammed, Mudasiru and Soyemi, Jumoke [2] use a meta-heuristic optimization (Firefly), this study compares the efficiency of Apriori against the FP-Growth algorithm.

The stability and reliability of interestingness measures are further discussed in the work of Al-Fayoumi, Mustafa and Alwidian, Jaber and Abusaif, Mohammad [3], which proposes an Intelligent Association Classification (IAC) technique for phishing detection. Similar to this study, Al-Fayoumi et al. [3] utilize the UCI Phishing Dataset and emphasize that standard Support and Confidence measures often fail to capture hidden patterns or deal with the estimation problems in high-dimensional data. To address this, they introduce the Harmonic Mean (HM) as a unified measure to rank and select association rules. Their results demonstrate that using a balanced statistical measure like HM allows the classifier to outperform traditional algorithms such as CMAR and FACA, achieving higher accuracy and precision.

There are strong methodological parallels between the study by Al-Fayoumi et al. [3] and this research, as both projects seek alternatives to the limitations of the support-confidence framework. However, the two studies focus on different aspects of rule evaluation. While Al-Fayoumi et al. focus on creating a single hybrid metric (Harmonic Mean) to build a more accurate classifier, this study provides a deeper investigation into the null-invariance property of multiple measures.

While Al-Fayoumi et al. use the Apriori algorithm as their base, this study expands the analysis by comparing it with FP-Growth to address the computational overhead of rule discovery in the UCI dataset.

The major difference of this study from other 3 studies is its focus on the "null-invariance" property. While the aforementioned studies primarily rely on one or two metrics like Support, Confidence, or Harmonic Mean to validate their findings, this research pivots toward a comprehensive analysis of eight different interestingness measures, including Lift, Chi-Square, All-Confidence, Max-Confidence, Cosine, and Kulczynski. This study introduces a controlled experimental setup by mathematically adding 10,000 synthetic null transactions to the dataset. This simulation allows for a rigorous test of null-invariance. By demonstrating how traditional metrics like Lift and Chi-Square can be misled by an increase in irrelevant transactions, this research underscores the necessity of using null-invariant measures to identify truly robust phishing patterns.

III. DATASET DESCRIPTION

The dataset used in this study is the "Phishing Websites Dataset," sourced from the UCI Machine Learning Repository. Dataset has 11,055 instances and 31 (30 predictive features and 1 target variable) attributes. All attributes are categorical, originally represented as numerical values: -1 (Phishing), 0 (Suspicious/Medium), and 1 (Legitimate). Some of the attributes in the dataset can be listed as follows: having_IP_Address, URL_Length, Shortining_Service, having_At_Symbol, double_slash_redirecting, having_Sub_Domain, SSLfinal_State, and Result. More details about attributes can be found at "Phishing Websites Dataset,"

The dataset does not have any missing value and shows a relatively balanced distribution. 55.7% (6,157 records) are Legitimate websites, and 44.3% (4,898 records) are Phishing websites. As shown in Figure 2, a Feature Correlation Heatmap was generated to visualize the global

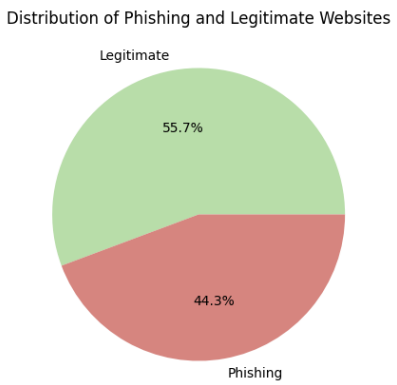

Fig. 1. Distribution of Phishing and Legitimate Websites.

relationships between attributes. This revealed that attributes such as `SSLfinal_State`, `URL_of_Anchor`, and `Prefix_Suffix` have the strongest linear relationship with the target class. The correlation analysis results, summarized

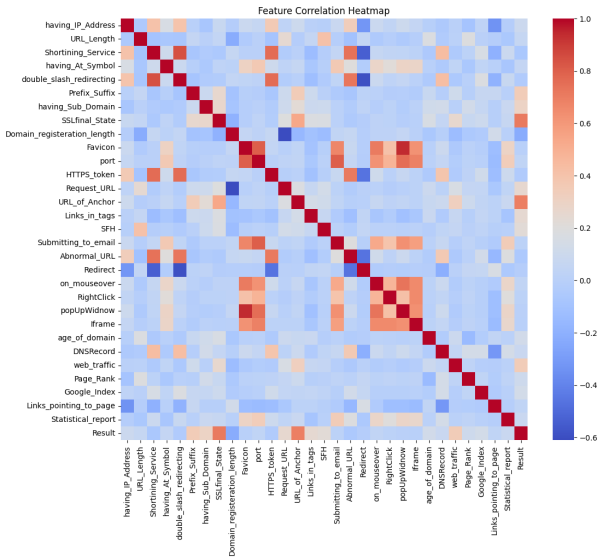

Fig. 2. Feature Correlation Heatmap.

in Table I, show the 20 features with the highest correlation scores.

IV. Methodology

A. Data Preprocessing

The raw dataset was provided in ARFF format with numerical encodings (-1, 0, 1). These values represent different security states of a website. However, such encodings are not intuitive when evaluating association rules. A semantic mapping process has been implemented to make the association rules more readable. Each numerical value was transformed into a human-readable string based on the technical definitions provided in the dataset's metadata. A sample of this mapping process is summarized in Table II. Full mapping process can be found at Appendix (Table VIII).

TABLE I
Correlation Coefficients of Features with the Target Variable (Result)

Feature Name	Correlation Score
SSLfinal_State	0.7147
URL_of_Anchor	0.6929
Prefix_Suffix	0.3486
web_traffic	0.3461
having_Sub_Domain	0.2983
Request_URL	0.2534
Links_in_tags	0.2482
Domain_registeration_length	0.2258
SFH	0.2214
Google_Index	0.1290
age_of_domain	0.1215
Page_Rank	0.1046
having_IP_Address	0.0942
Statistical_report	0.0799
DNSRecord	0.0757

TABLE II
Sample of Numerical-to-Semantic Feature Mapping

Feature Name	Original Value	Semantic Label
SSLfinal_State	-1 / 0 / 1	No_HTTPS / Suspicious / Trusted
URL_of_Anchor	-1 / 0 / 1	>67% Anchor / 31–67% / <31%
Prefix_Suffix	-1 / 1	Has_PrefixSuffix / No_PrefixSuffix
web_traffic	-1 / 0 / 1	Low / Medium / High_Traffic
Result	-1 / 1	Phishing / Legitimate

Association rule mining algorithms require a transactional binary format. After the mapping, the categorical features were transformed using one-hot encoding via the get_dummies method. This allows the algorithms to treat each attribute-value pair as a distinct item.

B. Algorithms

In this study, two algorithms have been selected: Apriori and FP-Growth.

Apriori is a classic breadth-first search algorithm that uses the downward closure property to prune the search space. Apriori requires multiple passes over the entire database, and the cost of candidate generation can be extremely high when the number of features or frequent patterns is large, as is often the case with cybersecurity datasets.

FP-Growth was developed to overcome the limitations of Apriori by adopting a depth-first, divide-and-conquer approach without explicit candidate generation. It transforms the dataset into a highly compact data structure called an FP-Tree (Frequent Pattern Tree). The database is scanned only twice. The first scan finds frequent items, and the second scan builds the FP-Tree, which stores compressed information about frequent patterns. By eliminating the need for candidate generation and reducing database scans to only two, FP-Growth is theoretically much faster and more scalable than Apriori, particularly for large and dense datasets where patterns are long and frequent.

1) Apriori and FP-Growth: In this study, the implementation of association rule mining was performed in several stages. Initially, the Apriori algorithm was applied to the entire

preprocessed dataset with a minimum support threshold of 0.15. However, due to the high dimensionality of the dataset, this approach encountered a Memory Error. Processing all possible feature combinations at this support level exceeded the available computational resources, demonstrating the curse of dimensionality inherent in large-scale association rule mining.

To overcome the memory bottleneck, a feature selection strategy was employed. Based on the Pearson correlation analysis performed in the previous stage, only the top 15 features with the highest absolute correlation coefficients relative to the target variable (Result) were selected for the mining process. Features can be found at Table I.

In order to observe the difference in speed between FP-Growth and Apriori, this method was also applied to the FP-Growth algorithm using the same minimum support and attributes.

From the frequent itemsets discovered at Apriori, association rules were generated using a minimum confidence threshold of 0.60. In order to provide a more comprehensive evaluation beyond the standard Support, Confidence, and Lift metrics, additional interestingness measures were implemented to analyze the rules from multiple perspectives. These metrics are Chi-Square, Cosine, All-Confidence, Max-Confidence and Kulczynski.

The data mining process generated a large number of rules. To focus on the results relating to phishing detection, the rules with a 'Result' value of 'Phishing' were selected and sorted in descending order by their 'Lift' value. The rules with a 'Result' value of 'Legitimate' were also selected and sorted the same way. Finally, rules for both "Phishing" and "Legitimate" classes were exported into separate CSV files.

2) FP-Growth on Entire Dataset: One more analysis with FP-Growth was performed using the entire dataset (all 30 original features). In this stage, one-hot encoding was applied to the complete dataset, which expanded the feature space to 70 attributes. Although the FP-Growth algorithm is generally efficient, processing such a high-dimensional dataset proved to be computationally demanding. To ensure system stability and optimize execution times, a constraint was implemented to limit the maximum length of frequent itemsets to 4. Consistent with the previous steps, a minimum support threshold of 0.15 and a minimum confidence threshold of 0.60 were maintained. Following the generation of the association rules, all eight evaluation metrics mentioned earlier were calculated. This comprehensive approach allowed for a deeper analysis of the relationships between features across the entire dataset. Following the calculation of these advanced metrics, the rules were again filtered for the Result=Phishing outcome and sorted by Lift.

3) Null Transactions: The final part of the methodology involves a controlled experiment to test the robustness of the evaluation metrics. In data mining, some metrics are sensitive to the total number of transactions, while others remain stable even if the number of null transactions increases.

To observe this behavior, the top five phishing rules were selected based on their initial Lift scores. The baseline metric values for the top five selected phishing rules were first recorded. Subsequently, a simulation was performed to evaluate the impact of increasing the dataset size without adding any new relevant information. Instead of physically generating synthetic data or identifying new rules, the total transaction count (N) was mathematically increased by 10,000, resulting in a new total of 21,055 records. All evaluation metrics were then recalculated using this updated value. This mathematical adjustment allowed for a clear observation of how each metric responds to a surge in null transactions while the absolute support (item counts) of the rules remained constant.

Given the potential for interestingness measures to provide biased results in the presence of null transactions, a final rule evaluation strategy was implemented. After performing the null-invariance test, the rules were re-ranked using the Kulczynski measure. Unlike Lift, the Kulczynski measure is null-invariant and provides a balanced assessment of the conditional probability of both the antecedent and the consequent, ensuring that the top rules reflect truly robust phishing signatures.

V. Results

A. Algorithmic Performance and Rule Discovery

In the first stage of the study, as described in the methodology section, both Apriori and FP-Growth algorithms were executed using the subset of 15 features to compare their performance. Although FP-Growth is typically expected to be faster than Apriori, the results for this specific dataset were counter-intuitive. According to the execution time comparison shown in Fig. 3, Apriori completed the task in 0.65 seconds, while FP-Growth took 91.22 seconds. The reason for this

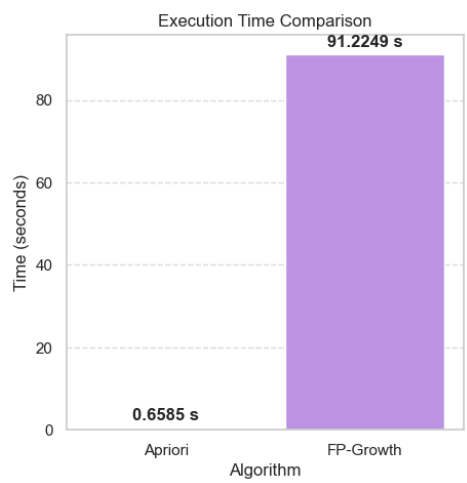

Fig. 3. Execution Time Comparison: Apriori vs. FP-Growth algorithms on the phishing dataset.

unexpected outcome could be that the candidate generation cost of Apriori was lower than the overhead required to build the complex FP-Tree structure of FP-Growth for this particular dataset and support threshold.

Following the extraction of frequent itemsets using the Apriori algorithm, association rules were generated, and various evaluation metrics were calculated. To identify the most

significant associations, the rules were sorted by their Lift values. The top 10 phishing-specific rules identified by the mining process are summarized in Table III. Only the most influential metrics are presented here; however, the comprehensive mathematical evaluation including all eight metrics is provided in Table IX in the Appendix section.

TABLE III
SUMMARY OF TOP 10 PHISHING DISCOVERY RULES (SORTED BY LIFT)

No	Antecedent (IF Website has...)	Conf.	Lift
1	Anchor > 67%, SSL No_HTTPS	1.000	2.257
2	Anchor > 67%, ReqURL High	1.000	2.257
3	PrefixSuffix, Anchor > 67%, SSL No_HTTPS	1.000	2.257
4	Anchor > 67%, GoogleIndexed, SSL No_HTTPS	1.000	2.257
5	Anchor > 67%, SFH Empty, SSL No_HTTPS	1.000	2.257
6	PrefixSuffix, Anchor > 67%, ReqURL High	1.000	2.257
7	PrefixSuffix, Anchor > 67%, SFH Empty, SSL No_HTTPS	1.000	2.257
8	PrefixSuffix, Anchor > 67%, GoogleIndexed, SSL No_HTTPS	1.000	2.257
9	PrefixSuffix, Anchor > 67%, DomainAge < 6M	0.997	2.250
10	Anchor > 67%, DomainAge < 6M	0.995	2.247

Each rule exhibits a Confidence level of nearly 100%, indicating that these feature combinations are almost certain indicators of phishing activity. As shown, the rule combinations involving `SSLfinal_State_No_HTTPS` and `URL_of_Anchor` greater than 67% yield the highest Lift scores.

In the second part of the rule discovery phase, rules leading to a Legitimate outcome were analyzed. The top 10 legitimate rules, sorted by their Lift values, are summarized in Table IV. Unlike phishing rules, legitimate patterns strongly rely on the presence of `SSLfinal_State_HTTPS_Trusted` and lower proportions of anchor URLs. Most rules in this category achieved a confidence level above 98%, proving to be highly reliable indicators of safe websites.

TABLE IV
SUMMARY OF TOP 10 LEGITIMATE DISCOVERY RULES (SORTED BY LIFT)

No	Antecedent (IF Website has...)	Conf.	Lift
1	Trusted_SSL, GoogleIndexed, Anchor < 31%	0.995	1.787
2	Trusted_SSL, Anchor < 31%	0.992	1.782
3	Trusted_SSL, StatReport_DontBelongTopPhishingIPs, Anchor < 31%	0.992	1.782
4	Trusted_SSL, GoogleIndexed, HighTraffic, NoSubDom	0.988	1.774
5	Trusted_SSL, StatReport_DontBelongTopPhishingIPs, HighTraffic, NoSubDom	0.987	1.772
6	Trusted_SSL, HighTraffic, NoSubDom	0.987	1.771
7	Trusted_SSL,Domain_ShortTerm, No_IP, ReqURL_Low, GoogleIndexed, HighTraffic	0.985	1.769
8	TagsBetween%17-81, Trusted_SSL, HighTraffic	0.982	1.764
9	Trusted_SSL, No_IP, ReqURL_Low, GoogleIndexed, HighTraffic	0.982	1.764
10	Trusted_SSL, Domain_ShortTerm, No_IP, ReqURL_Low, HighTraffic	0.981	1.762

Another stage of the analysis involved running the FP-Growth algorithm on the complete dataset with a constraint of `max_len=4`. This comprehensive mining process identified rules with perfect confidence (1.00), as summarized in Table V. These rules represent the most complex and robust "phishing signatures" discovered in this study. Consistent with the previous subset analysis, the combination of missing HTTPS and high-density anchor URLs remains the most significant predictor. A detailed table of these rules is provided in Appendix. (Table XI)

B. Null Transactions

The second part of the results focuses on the stability of evaluation metrics when the dataset size (N) is mathematically

TABLE V
TOP 10 PHISHING RULES FROM FULL DATASET (SORTED BY LIFT)

No	Antecedent (IF Website has...)	Conf.	Lift
1	Anchor > 67%, Redir 2-4, No_HTTPS	1.000	2.257
2	PrefixSuffix, Anchor > 67%, Links 1-2	1.000	2.257
3	RightClick, Anchor > 67%, Links 1-2	1.000	2.257
4	No_Iframe, No_HTTPS, Anchor > 67%	1.000	2.257
5	Anchor > 67%, SFH_Empty, No_HTTPS	1.000	2.257
6	Anchor > 67%, No_HTTPS	1.000	2.257
7	Anchor > 67%, No_HTTPS_Token, No_HTTPS	1.000	2.257
8	Anchor > 67%, No_HTTPS, SlashPos < 7	1.000	2.257
9	Anchor > 67%, LongURL, No_HTTPS	1.000	2.257
10	Anchor > 67%, AbnormalURL, No_HTTPS	1.000	2.257

increased. After selecting top 5 Phishing Rules from Full Dataset (Table V), every metric for these rules were recalculated. Lift and Chi-Square metrics showed extreme sensitivity to the dataset size. For all top rules, Lift increased by 90.45%, and Chi-Square surged by approximately 140%. This inflation happens because these formulas are dependent on the total transaction count (N). In contrast, the measures of Kulczyn-

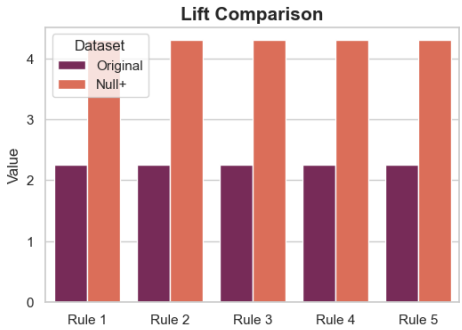

Fig. 4. Lift

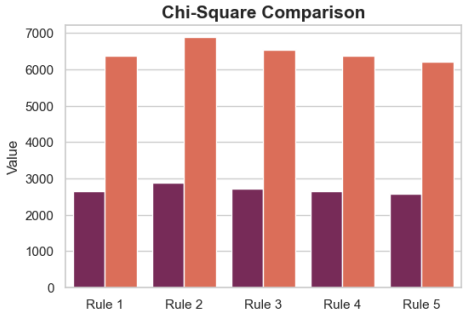

Fig. 5. Chi-Square

ski, Cosine, All-Confidence, and Max-Confidence remained 100% stable. This stability proves that these metrics only consider the presence of the features and the outcome, ignoring the irrelevant background data. For imbalanced data, null-invariant measures are significantly more reliable and robust than Lift or Chi-Square. A more detailed metric comparison can be found at Appendix (Table XII)

The results clearly illustrate the instability of the Lift and Chi-Square metrics. This sensitivity indicates that sorting by

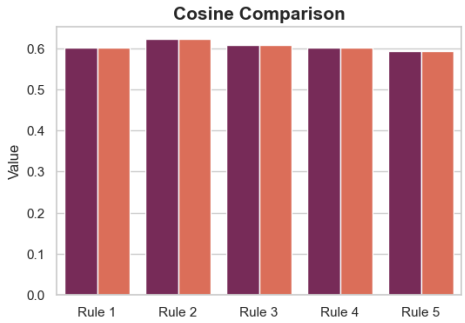

Fig. 6. Cosine

Fig. 7. All-Confidence

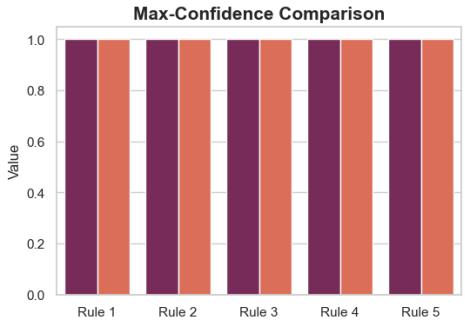

Fig. 8. Max-Confidence

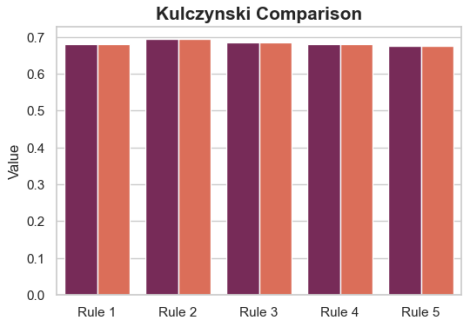

Fig. 9. Kulczynski

Lift can be misleading in growing or imbalanced datasets. Consequently, the top 10 phishing rules were re-evaluated and sorted based on the Kulczynski measure. These refined results are presented in Table VI (for the 15-feature subset) and Table VII (for the full dataset), providing a more reliable foundation for phishing discovery.

TABLE VI
TOP 10 PHISHING RULES SORTED BY KULCZYNSKI (15-FEATURE SUBSET)

No	Antecedent (IF Website has...)	Conf.	Lift	Kulc.
1	PrefixSuffix, Anchor > 67%	0.994	2.245	0.829
2	Anchor > 67%	0.989	2.232	0.826
3	PrefixSuffix, Anchor > 67%, SFH Empty	0.993	2.242	0.777
4	Anchor > 67%, SFH Empty	0.988	2.231	0.775
5	PrefixSuffix, Anchor > 67%, GoogleIdx	0.993	2.242	0.773
6	Anchor > 67%, GoogleIdx	0.987	2.227	0.770
7	PrefixSuffix, Anchor > 67%, PageRankLow	0.993	2.242	0.767
8	Anchor > 67%, PageRankLow	0.988	2.229	0.764
9	PrefixSuffix, Anchor > 67%, StatReport	0.995	2.245	0.761
10	Anchor > 67%, StatReport	0.988	2.230	0.758

TABLE VII
TOP 10 PHISHING RULES SORTED BY KULCZYNSKI MEASURE (FULL DATASET)

No	Antecedent (IF Website has...)	Conf.	Lift	Kulc.
1	PrefixSuffix, Anchor > 67%	0.994	2.245	0.829
2	Anchor > 67%	0.989	2.232	0.826
3	PrefixSuffix, Anchor > 67%, RightClick	0.994	2.244	0.809
4	RightClick, Anchor > 67%	0.988	2.231	0.806
5	PrefixSuffix, No_IFrame, Anchor > 67%	0.997	2.249	0.792
6	No_IFrame, Anchor > 67%	0.990	2.235	0.789
7	RightClick, No_IFrame, Anchor > 67%	0.990	2.235	0.789
8	PrefixSuffix, Anchor > 67%, Redirect 2-4	0.994	2.243	0.787
9	Anchor > 67%, Redirect 2-4	0.987	2.229	0.784
10	PrefixSuffix, Anchor > 67%, No_Shortening	0.994	2.243	0.782

VI. DISCUSSION

A. Algorithmic Performance: Apriori vs. FP-Growth

As observed in Results Section Fig. 3 , the execution time comparison between Apriori and FP-Growth yielded a counter-intuitive result where Apriori was significantly faster. In data mining theory, FP-Growth is generally expected to outperform Apriori. However, for this specific dataset at a minimum support threshold of 0.15, the overhead of constructing the complex FP-Tree structure was higher than Apriori's candidate generation process. This highlights that the choice of algorithm should be dependent on the dataset's density and the selected support threshold. Furthermore, the memory error encountered when all features were used without constraints confirms the curse of dimensionality, justifying the need for feature selection.

B. Phishing Indicators

The discovery of phishing signatures was initially conducted using the Lift measure as the primary ranking criterion. When comparing the top rules of the 15-feature subset (Table IX) with those of the full dataset (Table

XI) under the Lift ranking, a high degree of consistency was observed. In both cases, the rules with the highest confidence (1.00) and lift (2.257) were centered around the same core features: `SSLfinal_State_No_HTTPS` and `URL_of_Anchor_greater_than_67%`. However, the full dataset analysis introduced more behavioral nuances, such as `Redirect_between_2-4` and `Links_pointing_to_page_have_1_or_2`.

Despite the apparent strength of these rules, the simulation of null transactions (Table XII) revealed a critical flaw in the Lift and Chi-Square measures. These metrics proved to be highly sensitive to the total number of transactions (N), with Lift inflating by 90.45% despite the absolute support of the rules remaining constant. To ensure the reliability of the discovered patterns, a re-ranking strategy was implemented using the Kulczynski measure, which demonstrated 100% stability during the null-invariance test.

Re-evaluating the subset and full dataset rules using the Kulczynski measure led to significant shifts in rule priority.

1) Subset Analysis: Under Kulczynski ranking, the rule combining `Has_PrefixSuffix` and `Anchor > 67%` emerged as the most robust signature (Kulc: 0.829), followed closely by the singular Anchor $>$ 67% feature. This shift highlights that while many rules share identical Lift values, Kulczynski distinguishes those that have the most balanced conditional probability between the feature and the phishing outcome.

2) Full Dataset Analysis: The Kulczynski ranking on the full dataset yielded nearly identical top-tier rules as the subset, with Rule 1 again being the combination of `PrefixSuffix` and `Anchor > 67%`. However, a notable difference appeared in Rule 3 and Rule 4, where `RightClick_Enabled` superseded features like `SFH_Empty` found in the subset. This suggests that in a more complex feature space, behavioral manipulations like enabling right-click functions are more reliable indicators of a phishing attempt than form-handling errors.

Ultimately, these findings provide a deeper perspective on the initial Feature Correlation Heatmap (Fig. 2). While the heatmap identified `SSLfinal_State` (0.7147) as the feature with the strongest linear relationship, a finding initially confirmed by the top rules ranked by Lift, the transition to the Kulczynski measure reveals a critical shift in rule priority. Under the Kulczynski ranking, features such as `URL_of_Anchor` and `Prefix_Suffix` emerge as more prominent predictors, often superseding `SSLfinal_State` in the most robust signatures. This divergence effectively demonstrates the fundamental difference between the two metrics: while Lift is highly sensitive to the total transaction count (N), Kulczynski ensures a more balanced and stable evaluation of the conditional probabilities. This transition provides the necessary mathematical rigor to confirm that the identified signatures are not merely artifacts of dataset size, but are robust, null-invariant patterns inherent to phishing behavior.

VII. Conclusion and Future Work

This study provided a comprehensive evaluation of association rule mining algorithms and interestingness measures for the discovery of phishing website patterns. By comparing Apriori and FP-Growth, it was observed that while FP-Growth is generally superior in efficiency, the overhead of building an FP-Tree on a specific subset of features can make Apriori more performant for certain dataset densities.

The core contribution of this research lies in the critical analysis of interestingness measures. The experimental results demonstrated that standard metrics like Lift and Chi-Square are highly sensitive to the number of null transactions. In contrast, null-invariant measures provided the most stable and reliable ranking of rules. The analysis identified a high proportion of URL_of_Anchor as the most consistent and powerful indicator of phishing activity. These findings answer the research questions by confirming that null-invariant measures are essential for building trustworthy detection systems that remain robust regardless of the presence of legitimate traffic.

For future work, this study can be expanded in several directions. The phishing signatures discovered via association rules could be used as engineered features for supervised learning models to improve classification accuracy. Investigating how these phishing patterns evolve over time by applying the same methodology to more recent datasets would help in understanding the changing nature of cyberattacks.

References

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VIII. Appendix

GitHub link: Correlation and Rule Evaluation for Phishing Discovery

Table IX provides the complete set of evaluation metrics for the top 10 phishing rules. This comprehensive breakdown includes standard metrics (Support, Confidence, Lift), statistical significance (χ^2), and null-invariant measures (Cosine, All-Confidence, Max-Confidence, Kulczynski).

Table X presents the detailed statistical performance of the top 10 legitimate website discovery rules. This table includes all calculated metrics, allowing for a deep comparison of correlation strength and statistical independence.

Table XI provides the exhaustive evaluation of the top 10 phishing rules extracted from the entire dataset using the FP-Growth algorithm.

Table XII presents the precise numerical shift in evaluation metrics when the dataset is augmented with 10,000 synthetic null transactions.

TABLE VIII
EXACT MAPPING OF DATASET NUMERICAL VALUES TO SEMANTIC LABELS

Feature Name	Value	Semantic Label
having_IP_Address	-1 / 1	has_IP / no_IP
URL_Length	1 / 0 / -1	short_URL / medium_URL / long_URL
Shortining_Service	1 / -1	No_Shortining / Has_Shortining
having_At_Symbol	1 / -1	no_@_symbol / has_@_symbol
double_slash_redirecting	-1 / 1	last_position_bigger_than_7 / last_position_less_than_7
Prefix_Suffix	-1 / 1	Has_PrefixSuffix / No_PrefixSuffix
having_Sub_Domain	-1 / 0 / 1	Multi_SubDomain / One_SubDomain / No_SubDomain
SSLfinal_State	-1 / 1 / 0	No_HTTPS / HTTPS_Trusted / SSL_Suspicious
Domain_registeration_length	-1 / 1	Domain_ShortTerm / Domain_LongTerm
Favicon	1 / -1	Favicon_Internal / Favicon_External
port	1 / -1	Port_Standard / Port_NonStandard
HTTPS_token	-1 / 1	Has_HTTPS_Token_In_Domain / No_HTTPS_Token_In_Domain
Request_URL	1 / -1	ReqURL_LowExtContent / ReqURL_HighExtContent
URL_of_Anchor	-1 / 0 / 1	greater_than_%67_Anchor / between_%31_%67_Anchor / less_than_%31_Anchor
Links_in_tags	1 / -1 / 0	less_than_%17 / greater_than_%81 / between_%17_%81
SFH	-1 / 1 / 0	Empty_or_Blank / Same_Domain / Diiferent_than_Domain
Submitting_to_email	-1 / 1	using_mail_func / not_using_mail_func
Abnormal_URL	-1 / 1	no_hostname_in_URL / hostname_in_URL
Redirect	0 / 1	Redirect_between_2-4 / Redirect_Less_Than_1
on_mouseover	1 / -1	onMouseOver_doesnt_Changes_Status_Bar / onMouseOver_Changes_Status_Bar
RightClick	1 / -1	Enabled / Disabled
popUpWidnow	1 / -1	No_Popup_Text_Fields / Has_Popup_Text_Fields
Iframe	1 / -1	No_IFrame / Has_IFrame
age_of_domain	-1 / 1	Domain_Age_Less_6Month / Domain_Age_More_6Month
DNSRecord	-1 / 1	No_DNS_Record / Has_DNS_Record
web_traffic	-1 / 0 / 1	Low_Traffic / Medium_Traffic / High_Traffic
Page_Rank	-1 / 1	PageRank_Low / PageRank_High
Google_Index	1 / -1	Indexed by Google / Not Indexed by Google
Links_pointing_to_page	1 / 0 / -1	at least 2 external links / have 1 or 2 links / have no links
Statistical_report	-1 / 1	Belongs to Top Phishing IPs / Dont Belong to Top Phishing IPs
Result	-1 / 1	Phishing / Legitimate

TABLE IX
COMPLETE MATHEMATICAL EVALUATION OF TOP 10 PHISHING RULES

No	Antecedent (IF Website features...)	Sup.	Conf.	Lift	Cosine	All-C	Max-C	Kulc.	Chi-Sq
1	Anchor > 67%, SSL No_HTTPS	0.185	1.000	2.257	0.646	0.418	1.000	0.709	3161.6
2	Anchor > 67%, ReqURL High	0.165	1.000	2.257	0.611	0.373	1.000	0.686	2756.7
3	PrefixSuffix, Anchor > 67%, SSL No_HTTPS	0.185	1.000	2.257	0.646	0.418	1.000	0.709	3161.6
4	Anchor > 67%, GoogleIndexed, SSL No_HTTPS	0.153	1.000	2.257	0.588	0.346	1.000	0.673	2523.5
5	Anchor > 67%, SFH Empty, SSL No_HTTPS	0.156	1.000	2.257	0.593	0.352	1.000	0.676	2569.3
6	PrefixSuffix, Anchor > 67%, ReqURL High	0.165	1.000	2.257	0.611	0.373	1.000	0.686	2756.7
7	PrefixSuffix, Anchor > 67%, SFH Empty, SSL No_HTTPS	0.156	1.000	2.257	0.593	0.352	1.000	0.676	2569.3
8	PrefixSuffix, Anchor > 67%, GoogleIndexed, SSL No_HTTPS	0.153	1.000	2.257	0.588	0.346	1.000	0.673	2523.5
9	PrefixSuffix, Anchor > 67%, DomainAge < 6M	0.152	0.997	2.250	0.586	0.345	0.997	0.671	2491.7
10	Anchor > 67%, DomainAge < 6M	0.152	0.995	2.247	0.586	0.345	0.995	0.670	2484.6

TABLE X
COMPLETE MATHEMATICAL EVALUATION OF TOP 10 LEGITIMATE RULES

No	Antecedent (IF Website features...)	Sup.	Conf.	Lift	Cosine	All-C	Max-C	Kulc.	Chi-Sq
1	Trusted_SSL, GoogleIndexed, Anchor < 31%	0.170	0.995	1.787	0.552	0.306	0.995	0.651	1778.3
2	Trusted_SSL, Anchor < 31%	0.191	0.992	1.782	0.584	0.344	0.992	0.668	2031.9
3	Trusted_SSL, StatReport_DontBelongTopPhishingIPs, Anchor < 31%	0.165	0.992	1.782	0.543	0.297	0.992	0.645	1698.5
4	Trusted_SSL, GoogleIndexed, HighTraffic, NoSubDom	0.152	0.988	1.774	0.519	0.272	0.988	0.630	1511.1
5	Trusted_SSL, StatReport_DontBelongTopPhishingIPs, HighTraffic, NoSubDom	0.153	0.987	1.772	0.521	0.275	0.987	0.631	1521.6
6	Trusted_SSL, HighTraffic, NoSubDom	0.172	0.987	1.771	0.552	0.309	0.987	0.648	1748.7
7	Trusted_SSL, Dom-Short, No_IP, Req-Low, GoogleIdx, HighTraffic	0.162	0.985	1.769	0.535	0.291	0.985	0.638	1617.9
8	TagsBetween, Trusted_SSL, HighTraffic	0.151	0.982	1.764	0.517	0.272	0.982	0.627	1476.7
9	Trusted_SSL, No_IP, Req-Low, GoogleIdx, HighTraffic	0.170	0.982	1.764	0.547	0.305	0.982	0.644	1694.4
10	Trusted_SSL, Dom-Short, No_IP, Req-Low, HighTraffic	0.177	0.981	1.762	0.558	0.317	0.981	0.649	1771.9

TABLE XI
COMPLETE 8-METRIC EVALUATION OF TOP PHISHING RULES (FULL DATASET)

No	Antecedent (IF)	Sup.	Conf.	Lift	Cosine	All-C	Max-C	Kulc.	Chi-Sq
1	Anchor $>$ 67%, Redir 2-4, No_HTTPS	0.160	1.000	2.257	0.601	0.361	1.000	0.680	2652.6
2	PrefixSuffix, Anchor $>$ 67%, Links 1-2	0.171	1.000	2.257	0.622	0.387	1.000	0.693	2878.5
3	RightClick, Anchor $>$ 67%, Links 1-2	0.163	1.000	2.257	0.608	0.370	1.000	0.685	2726.1
4	No_Iframe, No_HTTPS, Anchor $>$ 67%	0.160	1.000	2.257	0.601	0.361	1.000	0.680	2652.6
5	Anchor $>$ 67%, SFH_Empty, No_HTTPS	0.156	1.000	2.257	0.593	0.352	1.000	0.676	2569.3
6	Anchor $>$ 67%, No_HTTPS	0.185	1.000	2.257	0.646	0.418	1.000	0.709	3161.6
7	Anchor $>$ 67%, Token_No, No_HTTPS	0.154	1.000	2.257	0.590	0.348	1.000	0.674	2539.3
8	Anchor $>$ 67%, No_HTTPS, SlashPos $<$ 7	0.159	1.000	2.257	0.599	0.359	1.000	0.679	2634.8
9	Anchor $>$ 67%, LongURL, No_HTTPS	0.151	1.000	2.257	0.584	0.341	1.000	0.670	2481.5
10	Anchor $>$ 67%, AbnormalURL, No_HTTPS	0.154	1.000	2.257	0.590	0.348	1.000	0.674	2541.1

TABLE XII
DETAILED METRIC STABILITY COMPARISON: ORIGINAL ($N \approx 11k$) VS. NULL+ ($N \approx 21k$)

Rule	Metric	Original (N=11k)	Null+ (N=21k)	Difference (%)
Rule 1	Lift	2.2570	4.2987	+90.45%
	Chi-Square	2652.7	6382.4	+140.60%
	Cosine	0.6015	0.6015	0.00%
	All-Confidence	0.3618	0.3618	0.00%
	Max-Confidence	1.0000	1.0000	0.00%
	Kulczynski	0.6809	0.6809	0.00%
Rule 2	Lift	2.2570	4.2987	+90.45%
	Chi-Square	2878.6	6877.2	+138.91%
	Cosine	0.6223	0.6223	0.00%
	All-Confidence	0.3873	0.3873	0.00%
	Max-Confidence	1.0000	1.0000	0.00%
	Kulczynski	0.6937	0.6937	0.00%
Rule 3	Lift	2.2570	4.2987	+90.45%
	Chi-Square	2726.1	6544.0	+140.05%
	Cosine	0.6084	0.6084	0.00%
	All-Confidence	0.3702	0.3702	0.00%
	Max-Confidence	1.0000	1.0000	0.00%
	Kulczynski	0.6851	0.6851	0.00%
Rule 4	Lift	2.2570	4.2987	+90.45%
	Chi-Square	2652.7	6382.4	+140.60%
	Cosine	0.6015	0.6015	0.00%
	All-Confidence	0.3618	0.3618	0.00%
	Max-Confidence	1.0000	1.0000	0.00%
	Kulczynski	0.6809	0.6809	0.00%
Rule 5	Lift	2.2570	4.2987	+90.45%
	Chi-Square	2569.3	6198.0	+141.23%
	Cosine	0.5935	0.5935	0.00%
	All-Confidence	0.3522	0.3522	0.00%
	Max-Confidence	1.0000	1.0000	0.00%
	Kulczynski	0.6761	0.6761	0.00%

TABLE XIII
COMPREHENSIVE EVALUATION METRICS FOR PHISHING SUBSET RULES (SORTED BY KULCZYNSKI)

No	Antecedent	Sup.	Conf.	Lift	Kulc.	Cosine	All-C	Max-C	Chi-Sq
1	PrefixSuffix, Anchor $>$ 67%	0.294	0.994	2.245	0.829	0.812	0.663	0.994	5707.2
2	Anchor $>$ 67%	0.294	0.989	2.232	0.826	0.810	0.663	0.989	5638.7
3	PrefixSuffix, Anchor $>$ 67%, SFH Empty	0.249	0.993	2.242	0.777	0.747	0.561	0.993	4531.7
4	Anchor $>$ 67%, SFH Empty	0.249	0.988	2.231	0.775	0.745	0.561	0.988	4479.8
5	PrefixSuffix, Anchor $>$ 67%, GoogleIdx	0.245	0.993	2.242	0.773	0.741	0.553	0.993	4441.0
6	Anchor $>$ 67%, GoogleIdx	0.245	0.987	2.227	0.770	0.739	0.553	0.987	4374.6
7	PrefixSuffix, Anchor $>$ 67%, PageRankLow	0.239	0.993	2.242	0.767	0.733	0.540	0.993	4305.5
8	Anchor $>$ 67%, PageRankLow	0.239	0.988	2.229	0.764	0.730	0.540	0.988	4250.3
9	PrefixSuffix, Anchor $>$ 67%, StatReport	0.234	0.995	2.245	0.761	0.724	0.527	0.995	4184.7
10	Anchor $>$ 67%, StatReport	0.234	0.988	2.230	0.758	0.722	0.527	0.988	4122.2

TABLE XIV
COMPREHENSIVE EVALUATION METRICS FOR PHISHING RULES (SORTED BY KULCZYNSKI)

No	**Antecedent**	**Sup.**	**Conf.**	**Lift**	**Kulc.**	**Cosine**	**All-C**	**Max-C**	**Chi-Sq**
1	PrefixSuffix, Anchor $>$ 67%	0.294	0.994	2.245	0.829	0.812	0.663	0.994	5707.2
2	Anchor $>$ 67%	0.294	0.989	2.232	0.826	0.810	0.663	0.989	5638.7
3	PrefixSuffix, Anchor $>$ 67%, RightClick	0.277	0.994	2.244	0.809	0.788	0.624	0.994	5245.3
4	RightClick, Anchor $>$ 67%	0.277	0.988	2.231	0.806	0.786	0.624	0.988	5177.5
5	PrefixSuffix, No_IFrame, Anchor $>$ 67%	0.260	0.997	2.249	0.792	0.765	0.588	0.997	4855.5
6	No_IFrame, Anchor $>$ 67%	0.260	0.990	2.235	0.789	0.763	0.588	0.990	4788.0
7	RightClick, No_IFrame, Anchor $>$ 67%	0.260	0.990	2.235	0.789	0.762	0.587	0.990	4778.8
8	PrefixSuffix, Anchor $>$ 67%, Redirect 2-4	0.257	0.994	2.243	0.787	0.759	0.580	0.994	4736.3
9	Anchor $>$ 67%, Redirect 2-4	0.257	0.987	2.229	0.784	0.757	0.580	0.987	4669.3
10	PrefixSuffix, Anchor $>$ 67%, No_Shortening	0.253	0.994	2.243	0.782	0.753	0.570	0.994	4630.1